CHM 221 – Fall 2026 Name_____
Prof. McDonald

Collaborators (if any)_____

Problem Set #5

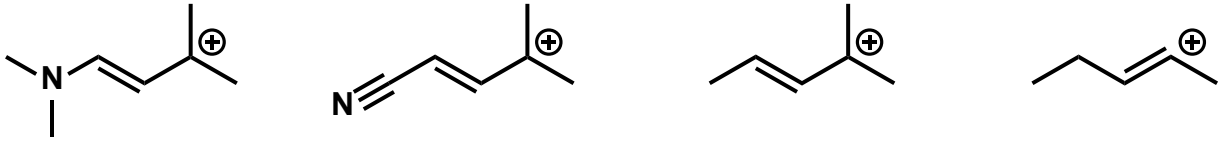
Due date: Monday, October 5thth by 9:00 AM

1. **Relative stability**
   Rank the following molecules based on their relative abilities to stabilize the depicted charge. (*1 = most stable)*.
   HINT: draw resonance structures/conjugate acids/bases to see how charges are delocalized!

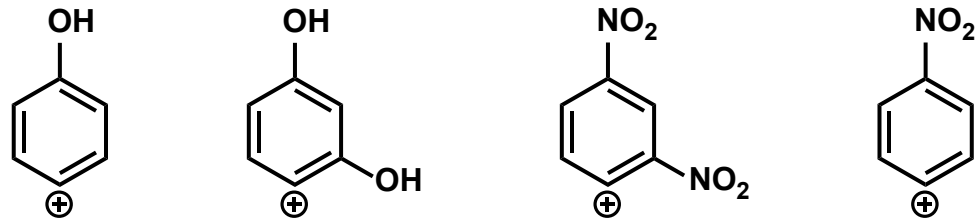
a)



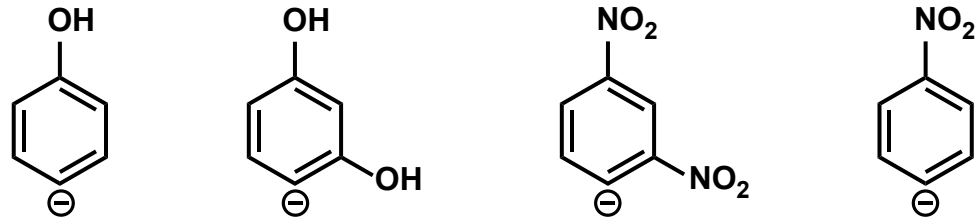
b)



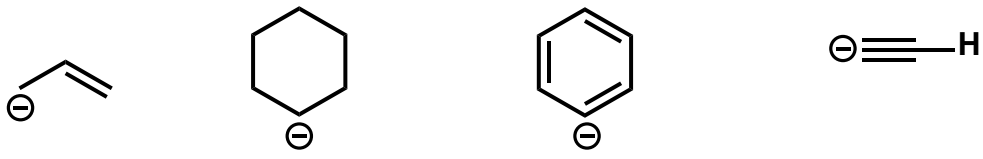
c)



d)



e)

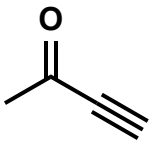


2. Sites of Acidic Protons

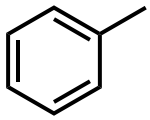
Circle the most acidic proton on each of the below molecules and indicate the appropriate pKa. Draw the hydrogen if it is implicit.

HINT: draw resonance structures of the conjugate bases to think about relative stabilities!

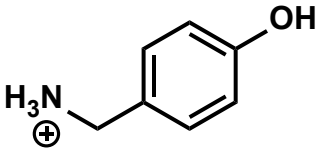
a.



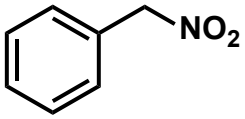
b.



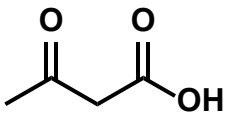
c.



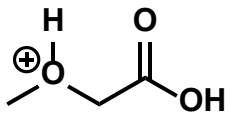
d.



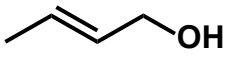
e.



f.



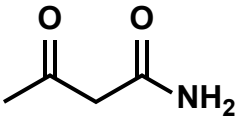
g.



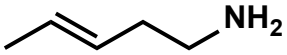
h.



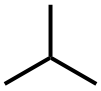
i.



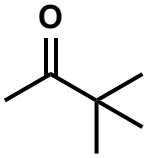
j.



k.



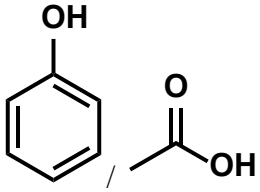
l.



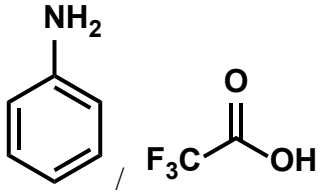
3. **Separation Schemes**
   You are in lab performing an acid/base separation of the below mixtures of compounds. Choose an appropriate acid or base to separate the mixture. Indicate which layer each compound would partition to.



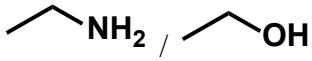
a. Phenol/Acetic Acid



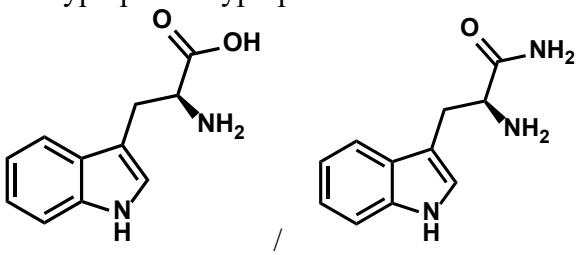
b. Aniline/Trifluoroacetic acid



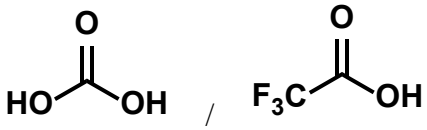
c. Ethylamine/Ethanol



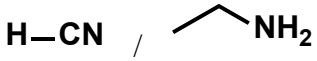
d. Tryptophan/Tryptophanamide



e. Carbonic acid/Trifluoroacetic acid



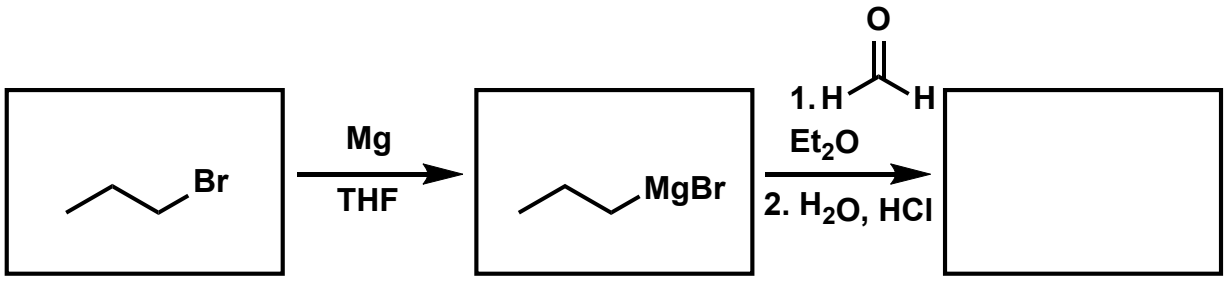
f. Hydrogen cyanide/Ethylamine



4. Complete the Reaction! (Organometallic Flavor)

Fill in the missing reagent, intermediate, or product for each reaction below.

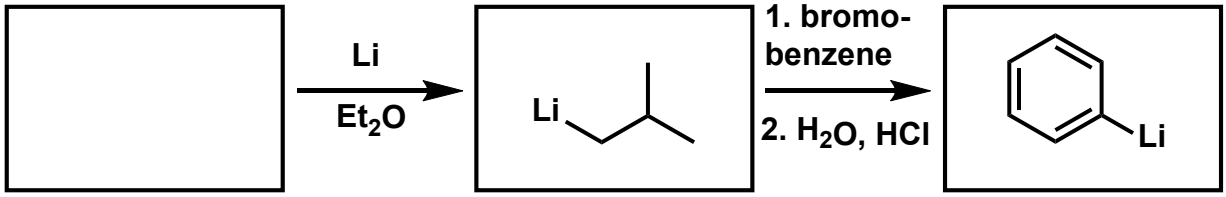
a.



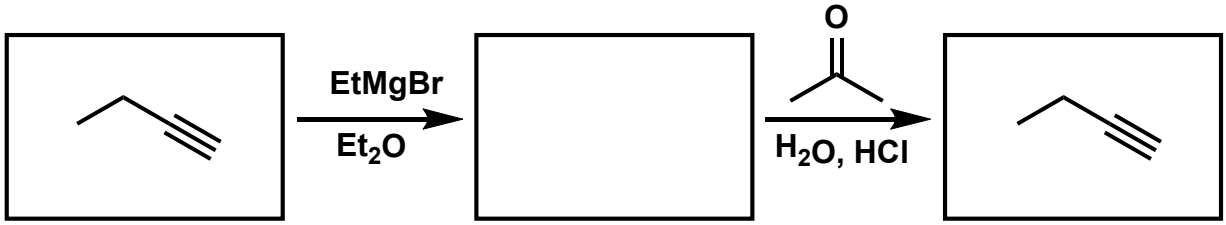
b.



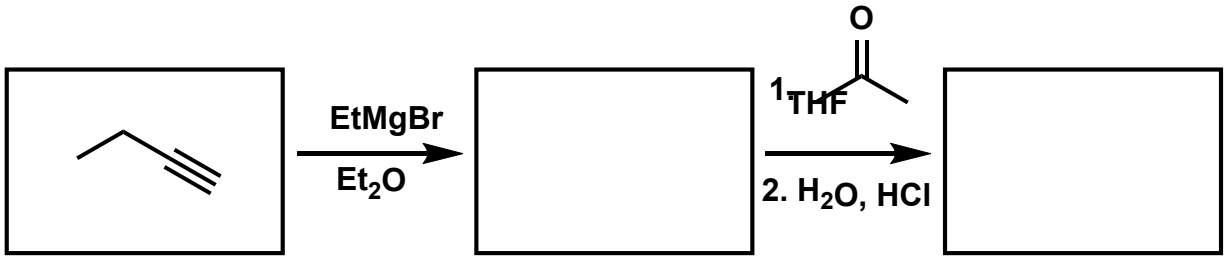
c.



d.

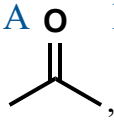
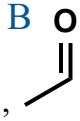
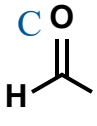
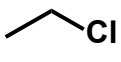
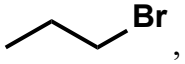
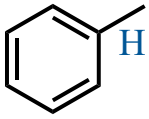
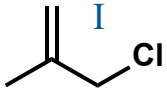


e.

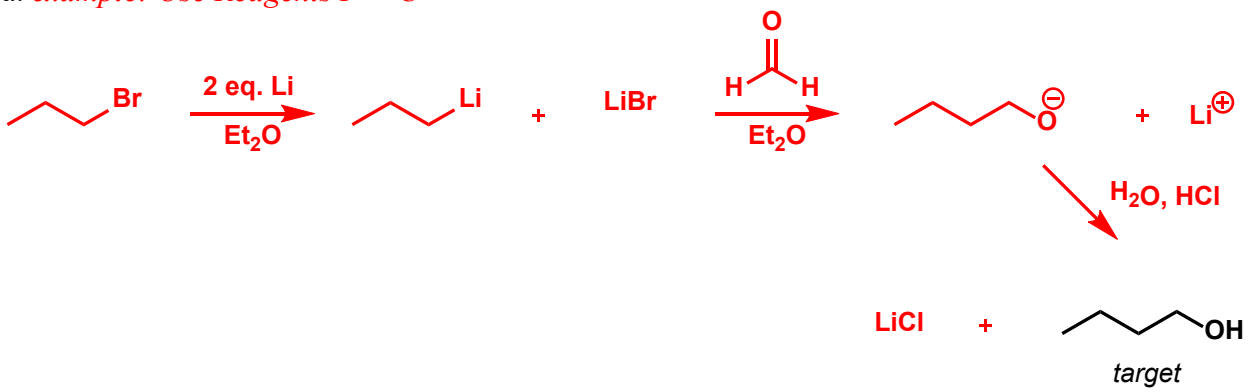


5. Build-A-Reaction

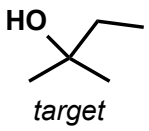
From the below pool of reagents and solvents, design reaction conditions that result in the indicated product.

Solvents	**Reagents**
THF, Et_2O H_2O, HCl	A  B  C  D  E  F 
Metals Li, Mg	G  H  I 

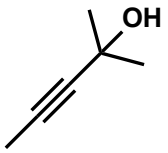
a. *example: Use Reagents F + C*



b.

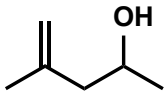


c.



target

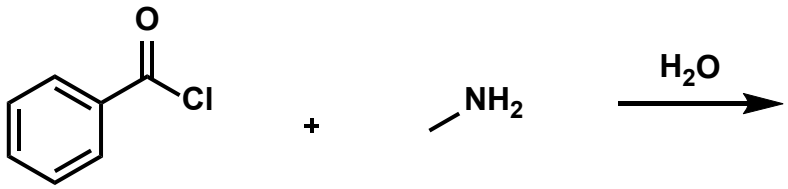
d.



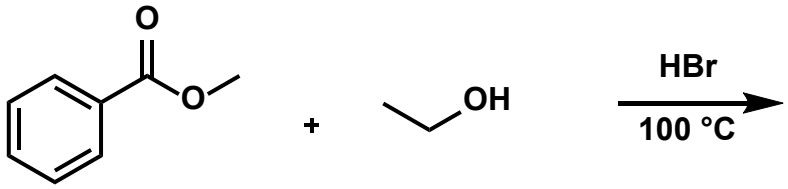
target

6. Predict the Products *(Extra! We may not get this far this week)*
Predict the resultant end product from the following results.

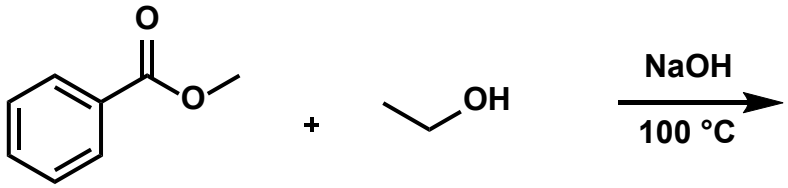
a.



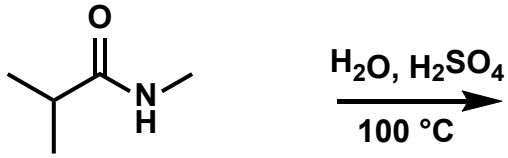
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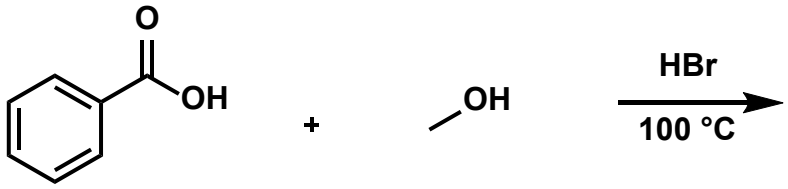
c.



d.



e.



f.

